

Network Information System

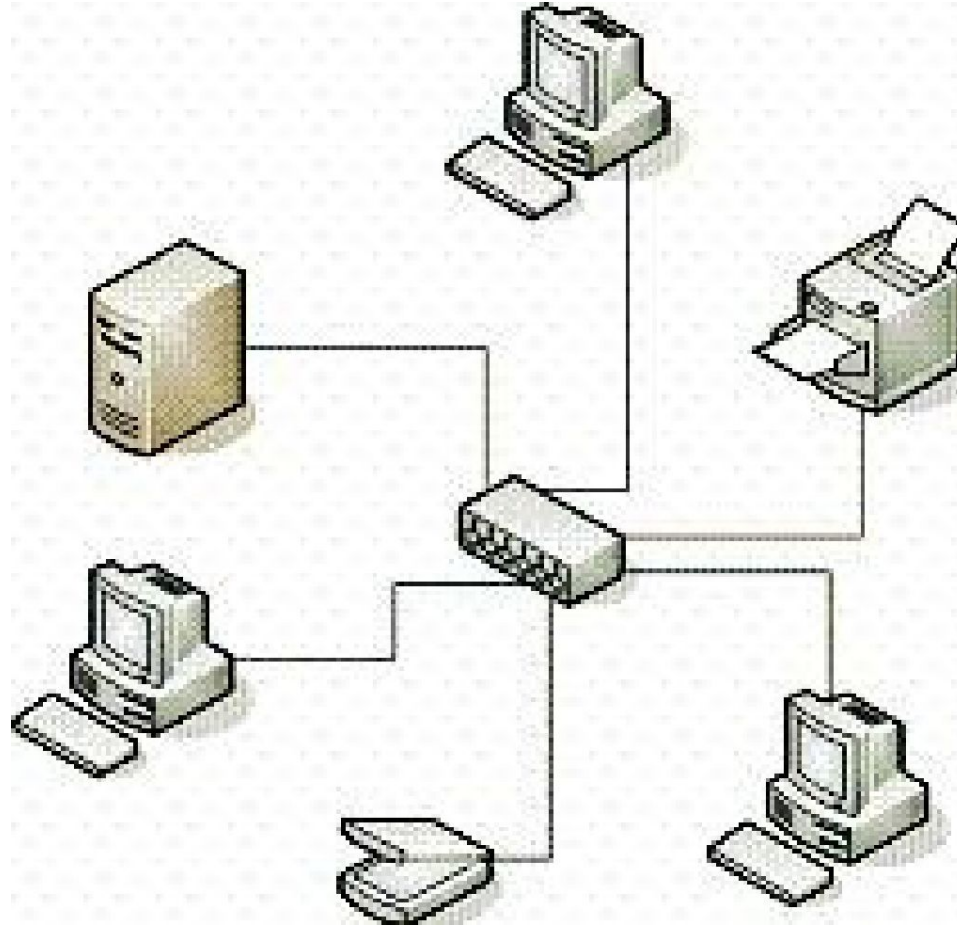
Computer Network

Computer Networks

- ▶ A network is a set of devices (also known as nodes, terminals or stations) interconnected by communication links.
- ▶ A computer network is a collection of computers and terminal devices connected together by a communication system to share resources and information.
- ▶ The set of computers may include large-scale computers, medium scale computers, mini computers and microprocessors.
- ▶ Computer networks, however, not only increase the reliability of computer resources and facilitate overall system developments, but also satisfy the primary objective of resource sharing which include device sharing, file sharing, program sharing, and program segmentation.

Example of a Computer Networks

Fig. 1 illustrates a computer network with devices like server, terminal, printer, hub and communication media etc.



Need and Scope of Networks

- ▶ Here are some of the ways a computer network can help the business:
- ▶ **File Sharing** - It provides sharing and grouping of data files over the network. The data file stored on the server allows easy access to all the staff in the organization.
- ▶ **Print Sharing** - It provides sharing of computer resources such as hard disk, printers etc by multiple users simultaneously to reduce the cost of installing and maintaining multiple resources in the organization.
- ▶ **E-Mail** - It provides collaboration tools such as group e-mail application which enables staff in the office to communicate with each other quickly and effectively.
- ▶ **Remote Access** - Network allows users to remotely access the data and information from organization's network via Internet in cost effective manner.
- ▶ **Shared Databases** - Network facilitates simultaneous access to the shared databases to multiple users at the same time by ensuring the integrity of the database.

Need and Scope of Networks

- ▶ **Fault Tolerance** - By using network, fault tolerance can be implemented as a defence against accidental data loss. Usually, primary and secondary line of defence backups the data in case of system failure. Additional measures can also be taken by attaching a server with uninterruptible power supply in case of power failure or blackouts.
- ▶ **Internet Access and Security** - It provides access to the Internet for transferring the document and to access the resources available on World Wide Web by maintain data security through firewall system in the organization's network.

Classifications of Networks

- ▶ Computer Networks can be classified in 4 classes
- ▶ **Class I : Function Based Classification**
 - Data Network: A communication network that transmits data.
 - Voice Network: A communication network that transmits voice.
 - Multimedia Network: A communication network that transmits data, voice, image, video etc.
- ▶ **Class II: Area Coverage Based Classification**
 - PAN : A personal area network (PAN) is a computer network used for data transmission among devices such as computers, telephones and personal digital assistants.
 - LAN: A Local Area Network (LAN) is a group of computers and network devices connected together, usually within the same building, campus or spanned over limited distance. It provides high speed data transfer and is relatively inexpensive.
 - MAN: A Metropolitan Area Network (MAN) is a larger network that usually spans in the same city or town. Cable network is an example of a MAN.
 - WAN A Wide Area Network (WAN) is not restricted to a geographical location, although it might be confined within the bounds of a state or country. The technology is high speed and relatively expensive. The Internet is an example of a world-wide public WAN.

Classifications of Networks

► Class III: Forwarding-based Classification

- Switched Network: Switched Network is a type of network that provide switched communication system and in which users are connected with each other through the circuits, packets, sometimes message switching and the control devices.
 - Active network elements like switch, router, gateways etc. participate in communication. Public switch telephone network is an example of switched networks.
- Shared Network A Shared Network is also known as Hubbed Network which is connected with a hub. When packets arrive in to the network, all segments can see packets. LAN using hub is an example of shared networks.

► Class IV: Ownership-based Classification

- Public Network: Network established for all users across the world is known as public network. Internet is an example of public network.
- Private Network Private Network is used by particular organization, particular campus or particular enterprise only. This is a network that is not available to the outside world. Intranet is an example of it.
- Virtual Private A Virtual Private Network (VPN) is a network that uses a public Network (VPN) network, such as the Internet, to provide secure access to organization's private network.
- Leased Network Dedicated or leased lines exist to support network communication.

Network Architecture

- ▶ Network architecture refers to the layout of the network, consisting of the hardware, software, connectivity, communication protocols and mode of transmission, such as wired or wireless.
- ▶ Every computer network supports two basic network architectures: **Client-Server** and **Peer-to-Peer**.

Client-Server

- ▶ Client-Server network consists of servers and clients. Servers are typically powerful computers running advanced network operating systems and user workstations (clients) which access data or run applications located on the servers.
- ▶ **Advantages:**
 - A client server can be scaled up to many services that can also be used by multiple users.
 - A client server enables the roles and responsibilities of a computing system. This means that it can update all the computers connected to it. An example of this would be software updates or hardware updates.
 - All the data is stored on the servers, which generally have far greater security controls than most clients. Servers can better control access and resources, to guarantee that only those clients with the appropriate permissions may access and change data.
 - The security is a lot more advanced than a peer to peer network.

Client-Server

► Disadvantages:

- When the server goes down or crashes, all the computers connected to it become unavailable to use.
- Simultaneous access to data and services by the user takes little more time for server to process the task.

Peer-to-Peer

- ▶ In Peer-to-Peer architecture, there are no dedicated servers. All computers are equal, and therefore, are termed as peer.
- ▶ The basic advantage of this architecture is simplicity of design and maintenance.
 - Since there is no server, all nodes on the network are fully employed and independent. Peer-to-peer networks are also known as workgroup.
 - No single computer holds the control of entire network.
 - Each system can install or upgrade their application/system software independently.
 - It is very simple and cost effective.
 - If one computer fails to work, all other computers connected to it continue to work.

Peer-to-Peer

- ▶ **Disadvantages:**

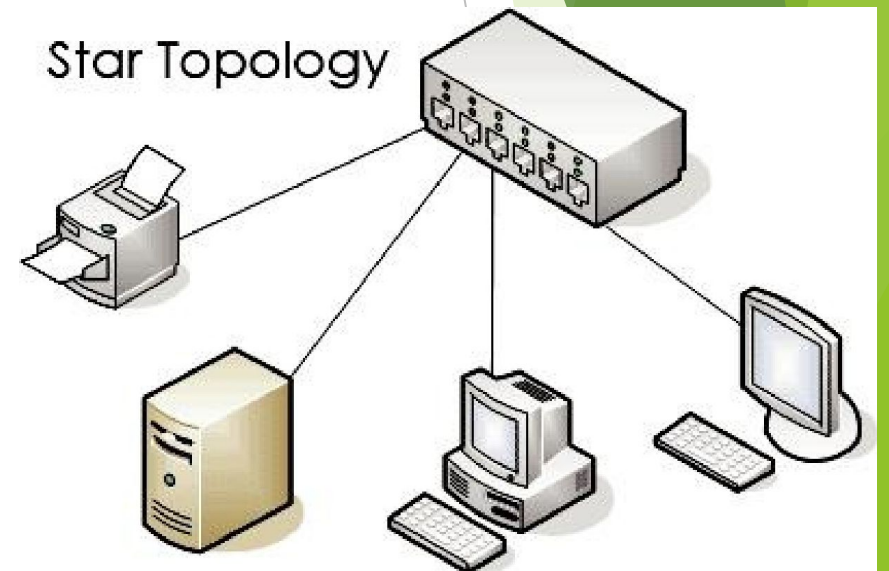
- ▶ There can be problem in accessing files if computers are not connected properly.
- ▶ It does not support connections with too many computers.
- ▶ The data security is very poor in this architecture.

Network Topologies

- ▶ The geometrical arrangement of computer resources, remote devices, and communication facilities is known as **Network Topology** or **Network Structure**.
- ▶ A computer network is comprised of nodes and links. A **node** is the end point of any branch in a computer, a terminal device, workstation or an interconnecting equipment facility.
- ▶ A link is a communication path between two nodes.
- ▶ A network structure determines which elements in a computer network can communicate with each other.
- ▶ Four basic network structures are discussed in this lecture

Star topology

- ▶ The most common structure or topology known as star network is characterized by communication channels emanating from centralized control
- ▶ The processing nodes in a star network topology interconnect directly with a central system.
- ▶ A star network is particularly appropriate for organisations that require a centralized data base or a centralized processing facility.
- ▶ For example, a star network may be used in banking for centralized record keeping in an on-line branch office environment.



Star Topology

► Advantages:

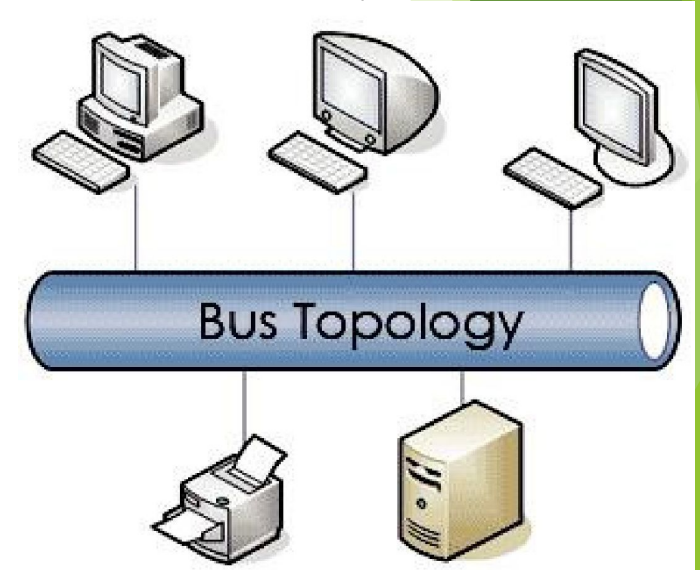
- It is easy to add new nodes and remove existing nodes.
- A node failure does not bring down the entire network.
- It is easier to diagnose network problems through a central hub.

► Disadvantages:

- If the central hub fails, the whole network ceases to function.
- It costs more to cable a star configuration than other topologies. (more cable is required than for a bus or ring configuration)

Bus topology

- ▶ In this structure or topology, a single network cable runs in the building or campus and all nodes are linked along with this communication line with two endpoints called the bus or backbone as shown in Fig.
- ▶ **Advantages:**
 - Reliable in very small networks as well as easy to use and understand.
 - Requires the least amount of cable to connect the computers together and therefore is less expensive than other cabling arrangements.
 - Is easy to extend. Two cables can be easily joined with a connector, making a longer cable for more computers to join the network.
 - A repeater can also be used to extend a bus configuration.



Bus topology

► Disadvantages

- Heavy network traffic can slow a bus considerably because any computer can transmit at anytime. But networks do not coordinate when information is sent. Computers interrupting each other can use a lot of bandwidth.
- Each connection between two cables weakens the electrical signal.
- The bus configuration can be difficult to trouble shoot. A cable break or malfunctioning computer can be difficult to find and can cause the whole network to stop functioning

Ring topology

- ▶ In this topology, the network cable passes from one node to another until all nodes are connected in the form of a loop or ring.
- ▶ There is a direct point-to-point link between two neighbouring nodes.
- ▶ These links are unidirectional which ensures that transmission by a node traverses the whole ring and comes back to the node, which made the transmission as shown in Fig..
- ▶ **Advantages:**
 - Ring networks offer high performance for a small number of workstations for larger networks where each station has a similar workload.
 - Ring networks can span longer distances than other types of network.
 - Ring networks are easily extendable.



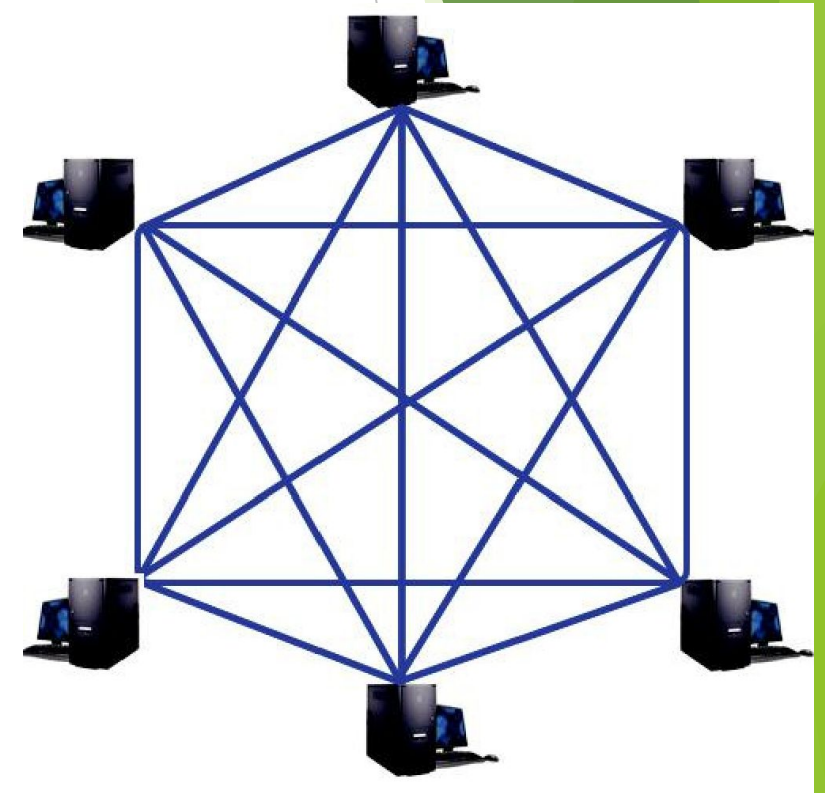
Ring Topology

► Disadvantages:

- Relatively expensive and difficult to install.
- Failure of one computer on the network can affect the whole network.
- It is difficult to trouble shoot a ring network.
- Adding or removing computers can disrupt the network

Mesh topology

- ▶ A mesh network may be fully connected (as shown in Fig..) or connected with only partial links.
- ▶ In fully interconnected topology, each node is connected by a dedicated point to point link to every node. This means that there is no need of any routing function as nodes are directly connected.
- ▶ The reliability is very high as there are always alternate paths available if direct link between two nodes is down or dysfunctional.
- ▶ Fully connected networks are not very common because of the high cost. Only military installations, which need high degree of redundancy, may have such networks, that too with a small number of nodes.
- ▶ Partially connected mesh topology is the general topology for wide area networks



Mesh Topology

► Advantages:

- Yields the greatest amount of redundancy in the event that one of the nodes fails where network traffic can be redirected to another node.
- Network problems are easier to diagnose.

▪ Disadvantages:

- The cost of installation and maintenance is high (more cable is required than any other configuration)

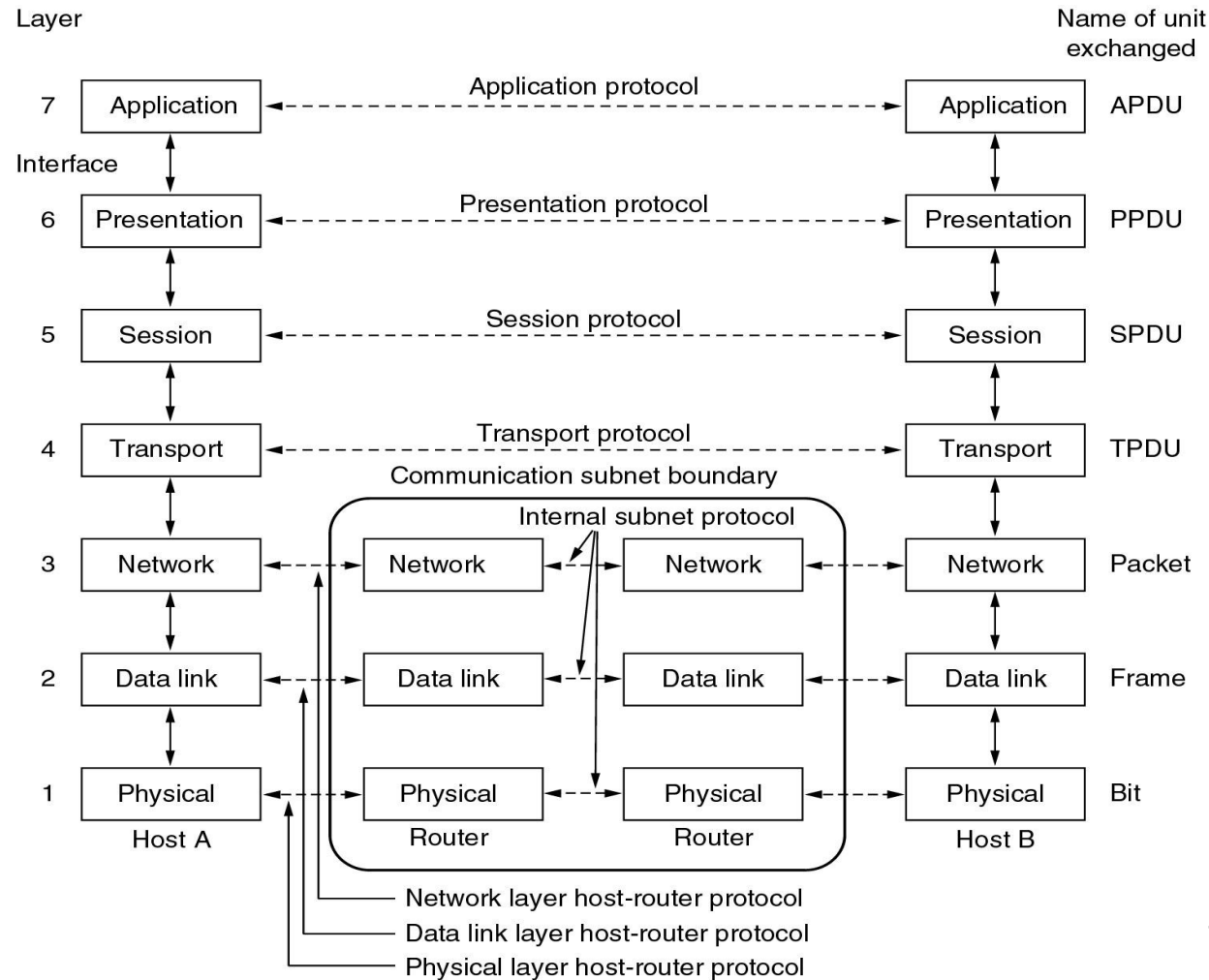
Reference Model

- The OSI Reference Model
- The TCP/IP Reference Model

Transmission Protocols

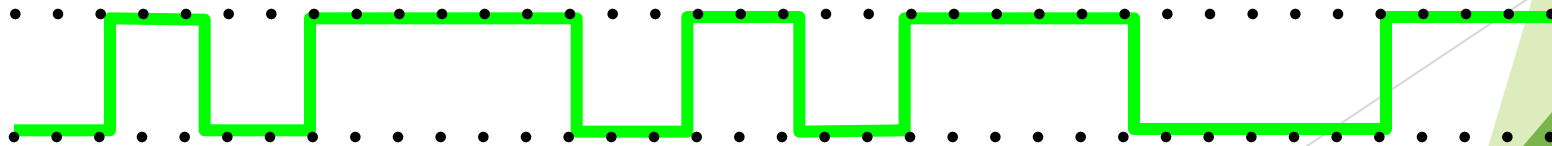
- ▶ Protocols are software that performs a variety of actions necessary for data transmission between computers.
- ▶ Stated more precisely, protocols are a set of rules for inter-computer communication that have been agreed upon and implemented by many vendors, users and standards bodies.
- ▶ Ideally, a protocols standard allows heterogeneous computers to talk to each other.
- ▶ There are two protocol stacks (groups protocols): OSI Model and TCP/IP Model

Open System Interconnection Model (OSI Model): OSI Model



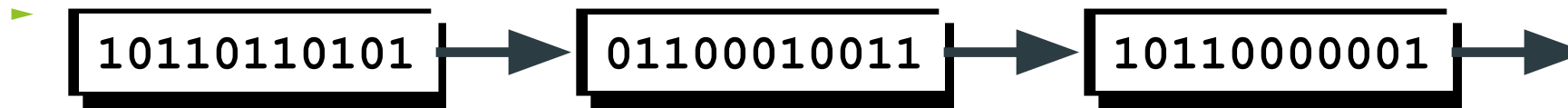
The Physical Layer

- Responsibility:
 - transmission of raw bits over a communication channel.
- Functions of the physical layer:
 - Moves bits between devices
 - Specifies voltage, wire speed and pin-out cables.
 - Determine how many pins the network connector has and what each pin is used for
- Issues
 - mechanical and electrical interfaces
 - time per bit
 - distances



The Data Link Layer - Data Link Control

- ▶ The main task of the data link layer is to take a raw transmission facility (a medium) and transform it into a line that appears free of undetected transmission errors to the network layer.
- ▶ To accomplish this, input data are broken into frames (typically a few hundred or a few thousand bytes).
- ▶ The frames are then transmitted sequentially and acknowledgment frames sent back by the receiver.
- ▶ The data link layer provides reliable transit of data across a physical network link.
- ▶ Different data link layer specifications define different network and protocol characteristics, including: physical addressing, network topology, error notification, sequencing of frames and flow control.



The Data Link Layer - Data Link Control

- ▶ Physical addressing (as opposed to networking addressing) defines how devices are addressed at the data link layer.
- ▶ Network topology consists of the data-link layer specifications that often define how devices are to be physically connected, such as in a bus, star or a ring topology.
- ▶ Error notification alerts upper-layer protocols that a transmission error has occurred.
- ▶ Sequencing of data frames reorders frames that are transmitted out of sequence.
- ▶ Flow control moderates the transmission of data so that the receiving device is not overwhelmed with more traffic than it can handle at one time.

The Data Link Layer - Data Link Control

- ▶ Functions of the data link layer:
 - ▶ Combines packets into bytes and then bytes into frames.
 - ▶ Create and to recognize frame boundaries - typically by attaching special bit patterns to the beginning and end of the frame.
 - ▶ Provides access to media using MAC address*.
 - ▶ Performs error detection (not correction!).
- ▶ *MAC Address (Hardware address/Ethernet address/Physical address) □
 - ▶ MAC is short for Media Access Control.
 - ▶ The MAC address is the hardware address that uniquely identifies a network device or a node (normally a Network Interface Card, called NIC) on the network.
 - ▶ Every NIC has a UNIQUE MAC address.
 - ▶ Note: A MAC Address is not the same thing as an IP address!
 - ▶ The address currently has the format: xx-xx-xx-xx-xx-xx. The first 3 octets of the address is the manufacturer's Organizationally Unique Identifier (OUI).

The Network Layer

- ▶ The main task of the network layer is to determine how data can be delivered from source to destination, i.e., provides routing and related functions that enable multiple data links to be combined into an internetwork.
- ▶ This is accomplished by the logical addressing (IP address as opposed to the physical addressing - MAC address) of devices.
- ▶ The network layer supports both connection-orientated and connectionless service from higher-layer protocols.
- ▶ Network-layer protocols typically are routing protocols, but other types of protocols are implemented at the network layer as well.

The Network Layer

- ▶ Functions:
 - ▶ Path selection between end-systems (routing).
 - ▶ Subnet flow control.
 - ▶ Fragmentation & reassembly
 - ▶ Provides logical addressing, which routers use for path determination.
 - ▶ To control congestion.
 - ▶ To do accounting.
 - ▶ To allow interconnection of heterogeneous networks (translation between different network types.)

The Transport Layer

- ▶ Responsible for packet handling.
 - ▶ Ensures error free delivery.
 - ▶ Repackages messages, divides messages into smaller packets (fragments and reassembles data) and handles error handling.
 - ▶ Ensures proper sequencing and without loss and duplication.
 - ▶ Takes action to correct faulty transmissions.
 - ▶ Acknowledges successful receipt of data.

Functions of the Transport Layer

- ▶ Flow control.
 - ▶ Flow control manages data management between devices so that the transmitting device does not send more data than the receiving device can process.
- ▶ Multiplexing.
 - ▶ Multiplexing enables data from several applications to be transmitted onto a single physical link.
- ▶ Virtual circuit management.
 - ▶ Virtual circuits are established, maintained, and terminated by the transport layer.
- ▶ Error checking and recovery.
 - ▶ Error checking involves creating various mechanisms for detecting transmission errors, while error recovery involves taking an action, such as requesting that data be retransmitted, to resolve any errors that occur

The Session Layer

- ▶ The session layer allows users on different machines to establish sessions between them.
- ▶ A session allows enhanced services useful in some applications.
- ▶ The session layer establishes, manages and terminates communication sessions between presentation layer entities.
- ▶ Communication sessions consist of service requests and service responses that occur between applications located in different network devices. These requests and responses are coordinated by protocols implemented at the session layer.

Functions of the session layer:

- ▶ Dialog control - session can allow traffic to go in both directions at the same time or in only one direction at a time. If traffic can go only in one way at a time, the session layer can help to keep track of whose turn it is.
- ▶ Token management - for some protocols it is essential that both sides do not attempt the same operation at the same time. The session layer provides tokens that can be exchanged. Only the side holding the token may perform the critical action.
- ▶ Synchronization - by inserting checkpoints into the data stream the layer eliminates problems with potential crashes at long operations. After a crash, only the data transferred after the last checkpoint have to be repeated.

The Presentation Layer

- ▶ Responsibilities:
 - ▶ data encryption
 - ▶ data compression
 - ▶ data conversion
- ▶ Functions of the presentation layer:
 - ▶ It is concerned with the syntax and semantics of the information transmitted.
 - ▶ Presents data.
 - ▶ Handles processing such as encryption etc

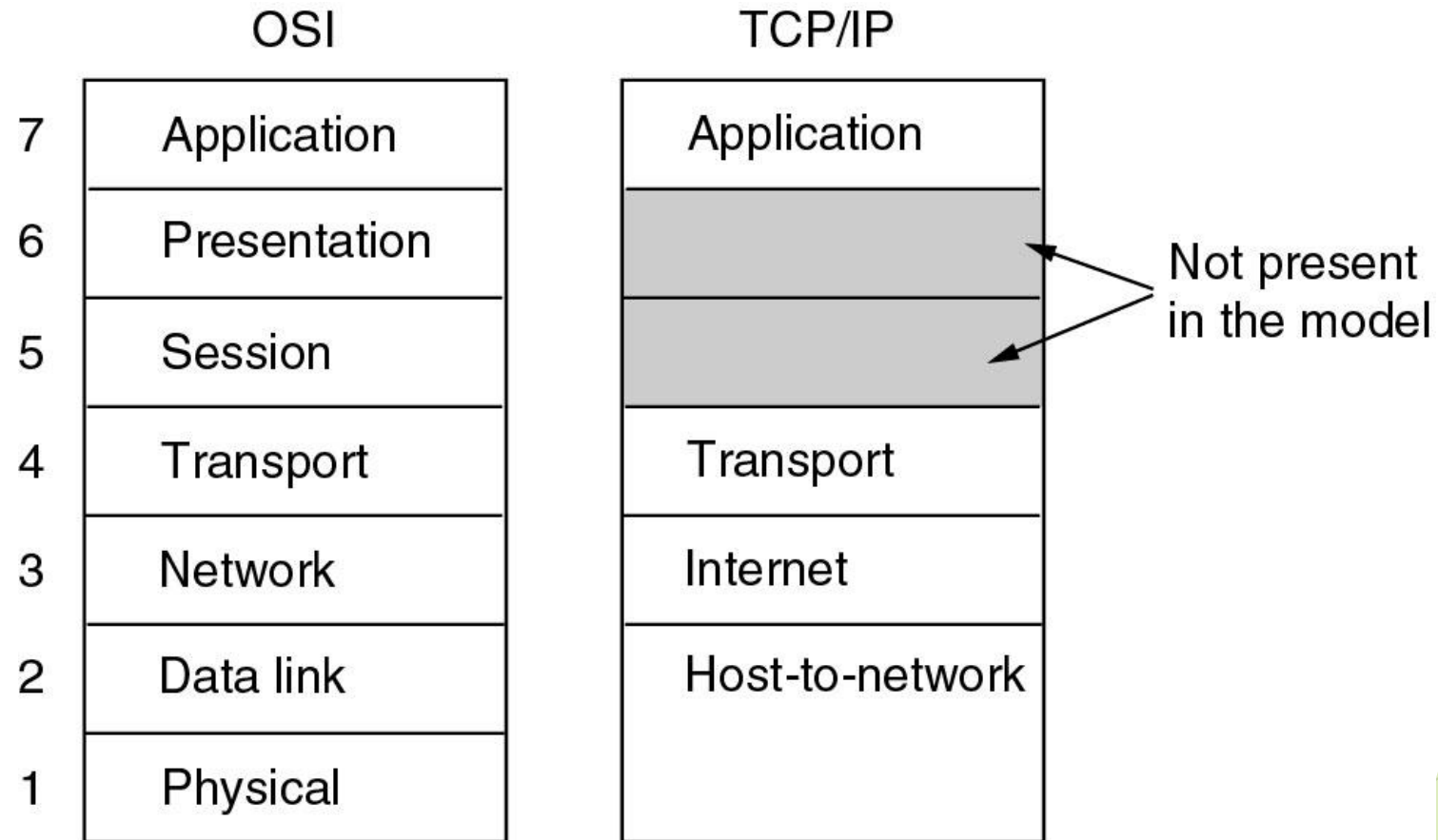
The Application Layer

- ▶ Responsibilities:
 - ▶ anything not provided by any of the other layers
- ▶ Issues:
 - ▶ application level protocols
 - ▶ appropriate selection of “type of service”

Summary of OSI RM

LAYERS	FUNCTIONS	CORRESPONDING PROCOTİLS
Application	Provide services to applications	HTTP, SMTP, FTP, NFS, Telnet, SMB
Presentation	Formatting, Compression, Encryption	JPEG, MIDI, MPEG etc etc
Session	Data transfer, class of service, control data exchange	Network File System (NFS), SQL, RPC
Transport	Quality and reliability, ensures data received, segments	TCP, UDP, SPX, NetBEUI
Network	Path selection, logical addressing, routing	IP, IPX, RIP, ICMP, ARP, RARP,
Data Link	Reliable data transfer across media; physical addressing	OSPF, NetBEUI, DLC, HDLC, SLIP, PPP
Physical	Transmit data on media	NONE

TCP/IP Reference Model



Protocols in TCP/IP model

